



D Y PATIL
COLLEGE OF
ENGINEERING & TECHNOLOGY
(AN AUTONOMOUS INSTITUTE)
KASABA BAWADA, KOLHAPUR



Google Developer Group
On Campus DYP CET

Techsprint

DimensionX Hackathon

WHERE
STRANGE IDEAS
TAKE SHAPE

Team Details

- Team name: Team NOVA
- Team leader name: Maithili Sanjay Raut
- Problem Statement:

Brief about your solution and problem statement addressing

Problem -

- **Flat Learning:** Traditional textbooks fail to convey the 3D complexity of the human body, which prevents true spatial understanding.
- **Cost & Access:** Expensive cadaver labs are out of reach for many institutions, limiting student hands-on experience.
- **Engagement Gap:** Static diagrams lead to disengagement among younger students who are used to interactive technology.

Solution -

- **3D Mastery:** Replaces flat images with 360 models for true anatomical depth.
- **Virtual Lab:** Provides digital cadaver labs on any device to eliminate high physical costs.
- **Adaptive AI:** Scales learning complexity from Grade 9 to professional surgical levels.
- **AR/VR Immersion:** Uses gamified Augmented Reality to maximize long-term memory retention.



Opportunities

a. How different is it from any of the other existing ideas?

1. Immersive AR: Unlike static sites, we provide instant AR/VR links to place 3D organs on your desk for hands-on spatial learning.
2. Tiered AI: Most platforms are "one-size-fits-all," but our system scales complexity from Grade 9 basics to professional surgical depth.
3. Active Engagement: We shift from passive video watching to gamified active recall, making learning interactive rather than just observational.

b. How will it be able to solve the problem?

1. Access: Turns smartphones into "Virtual Labs," eliminating million-dollar facility costs.
2. Depth: Replaces 2D diagrams with 360 models for true spatial understanding.
3. Interest: Uses AR and gamification to boost student engagement and memory retention.

List of features offered by the solution

1. AI Assistance -

- **Smart Guidance:** Real-time AI analysis that assists surgeons during procedures by identifying anatomical landmarks.
- **Decision Support:** Provides data-driven insights to help the surgical team make informed choices under pressure.

2. 3D Model View -

- **Spatial Understanding:** High-resolution 3D reconstructions of patient anatomy from CT or MRI scans.
- **Pre-Surgical Planning:** Allows surgeons to rotate, zoom, and "dissect" a digital model before entering the operating room.

3. AR/VR View -

- **Immersive Training:** Virtual Reality (VR) environments for risk-free surgical practice.
- **Augmented Reality (AR):** Overlays digital information directly onto the patient's body during surgery to improve precision.

4. Interactive Quizzes -

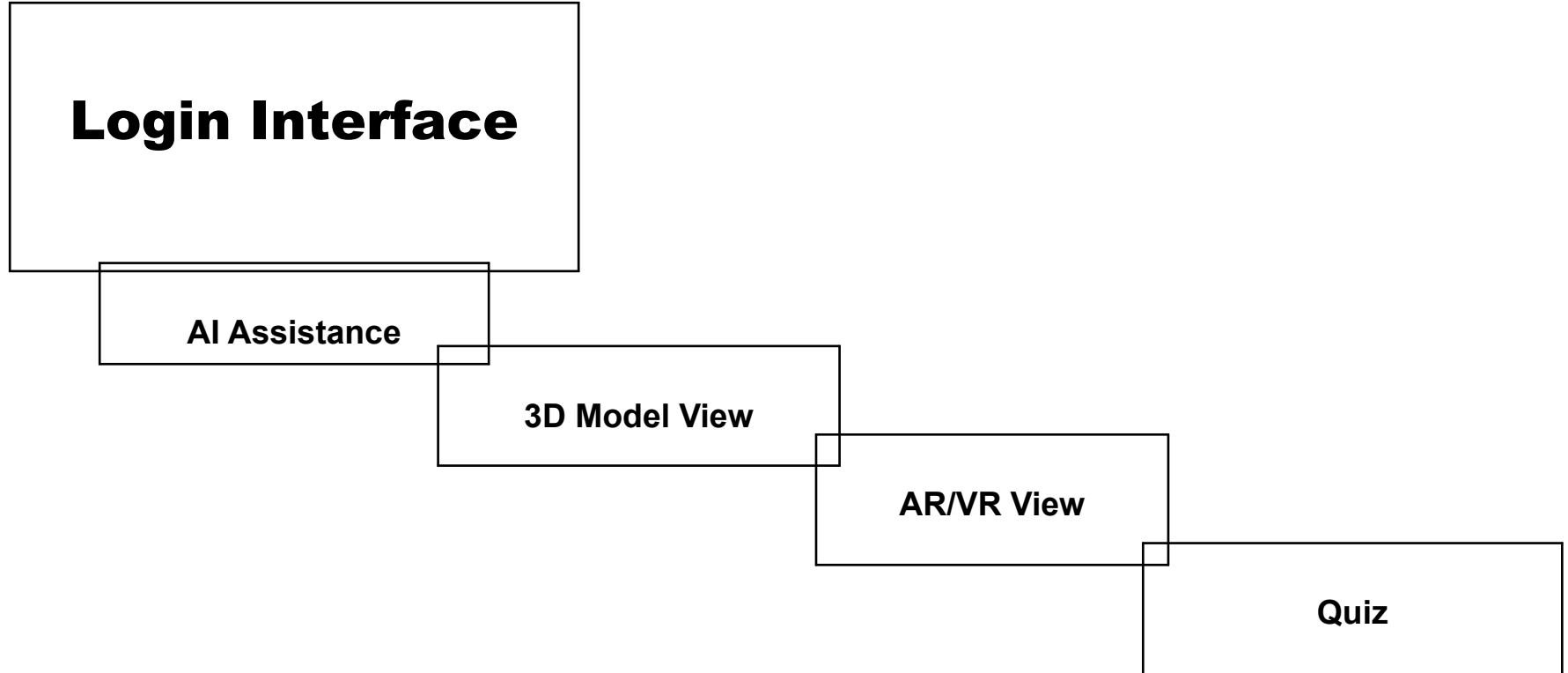
- **Skill Assessment:** Gamified "Quiz Time" modules to test medical knowledge and surgical protocols.
- **Progress Tracking:** Provides instant feedback and performance metrics for students and residents.

Google Technologies used in the solution

- **Gemini 1.5 Flash:** Generates surgical flashcards and anatomical data.
- **Google Cloud Storage:** Scalable object storage for medical documents, images, and 3D assets.
- **Google 3D Viewer:** Enables mobile-friendly viewing and AR capabilities.
- **Firebase:** Manages secure user login and Google Forms integration.



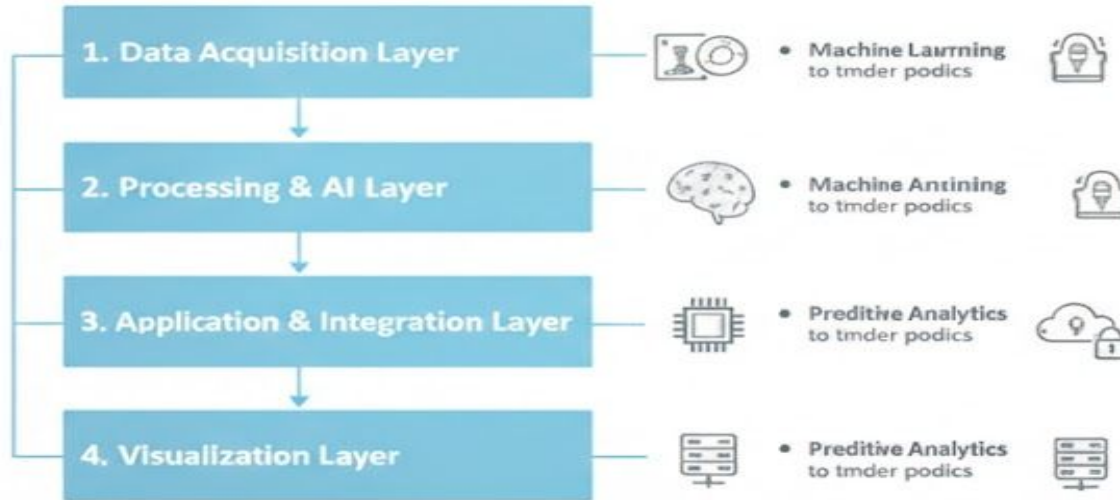
Process flow diagram or Use-case diagram





Architecture diagram of the proposed solution

System Architecture of SurgicalAI



Snapshots of the MVP



Additional Details/Future Development (if any)

MRI & CT Scan Integration: Implementing a feature to upload real patient DICOM files, automatically converting 2D scans into interactive 3D "Digital Twins" for pre-surgical planning.

Collaborative Multiplayer Labs:

Developing "Virtual Operating Rooms" where a teacher and multiple students can join the same 3D space from different locations for remote practical sessions.

Cross-Platform Expansion:

Launching dedicated mobile applications with advanced AR capabilities for offline learning in rural or low-connectivity educational areas.

Provide links to your:

1. **GitHub Public Repository :**
2. **Demo Video Link (3 Minutes) :**
3. **MVP Link :**



Thank you!